

Hospital Management System

Hospital Management System

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Introduction

The Hospital Management System (HMS) is a digital software solution designed to manage and streamline hospital operations. It automates patient registration, appointment scheduling, doctor management, pharmacy operations, laboratory workflows, billing, and medical records. The system reduces manual work, minimizes errors, and improves efficiency.

1. Requirements

2.1 Functional Requirements **Admin**

- Add, update, and remove hospital staff
- Manage departments and roles- View overall system reports

Receptionist

- Register new patients
- Schedule, update, cancel appointments- Manage check-in and check-out

Doctor

- View patient history
- Add diagnosis and treatment
- Prescribe medicines- Request lab tests

Pharmacist

- Manage medicine inventory
- Issue medicines
- Update stock levels

Lab Technician

- View test requests
- Conduct tests
- Upload results

Patient

- View appointments
- Access prescriptions
- Download reports

2.2 Non-Functional Requirements - Security: encrypted passwords, role-based access

- Performance: fast response time
- Usability: simple interface
- Scalability: expandable system
- Reliability: 24/7 uptime
- Data Integrity: consistent and accurate data

3. Use Cases and Flow of Processes

3.1 Actors - Admin

- Receptionist
- Doctor
- Patient
- Pharmacist
- Lab Technician

3.2 Major Use Cases

1. Register Patient

Actor: Receptionist

Description: Enter patient details and generate ID.

2. Schedule Appointment

Actor: Receptionist / Patient

Description: Book appointment based on doctor availability.

3. Doctor Consultation

Actor: Doctor

Description: View patient record, add diagnosis, prescribe medicines or tests.

4. Laboratory Test

Actor: Lab Technician

Description: Conduct test and upload result.

5. Pharmacy Operation

Actor: Pharmacist

Description: Issue medicines and update inventory.

6. Billing

Actor: Accounts/Admin

Description: Generate final invoice.

3.3 Flow of Processes 1. Patient Registration → System generates patient ID

2. Appointment Booking → Check doctor availability

3. Consultation → Doctor adds diagnosis

4. Lab Test → Technician performs and uploads report

5. Pharmacy → Prescription is issued

6. Billing → Total charges are calculated

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2. Sommerville, I. Software Engineering (10th Edition).

3. IEEE SRS Documentation Standards.

4. Research papers on Hospital Information Systems.

5. Official documentation (MySQL, PHP, Python, etc.).